Spike Training Decoding

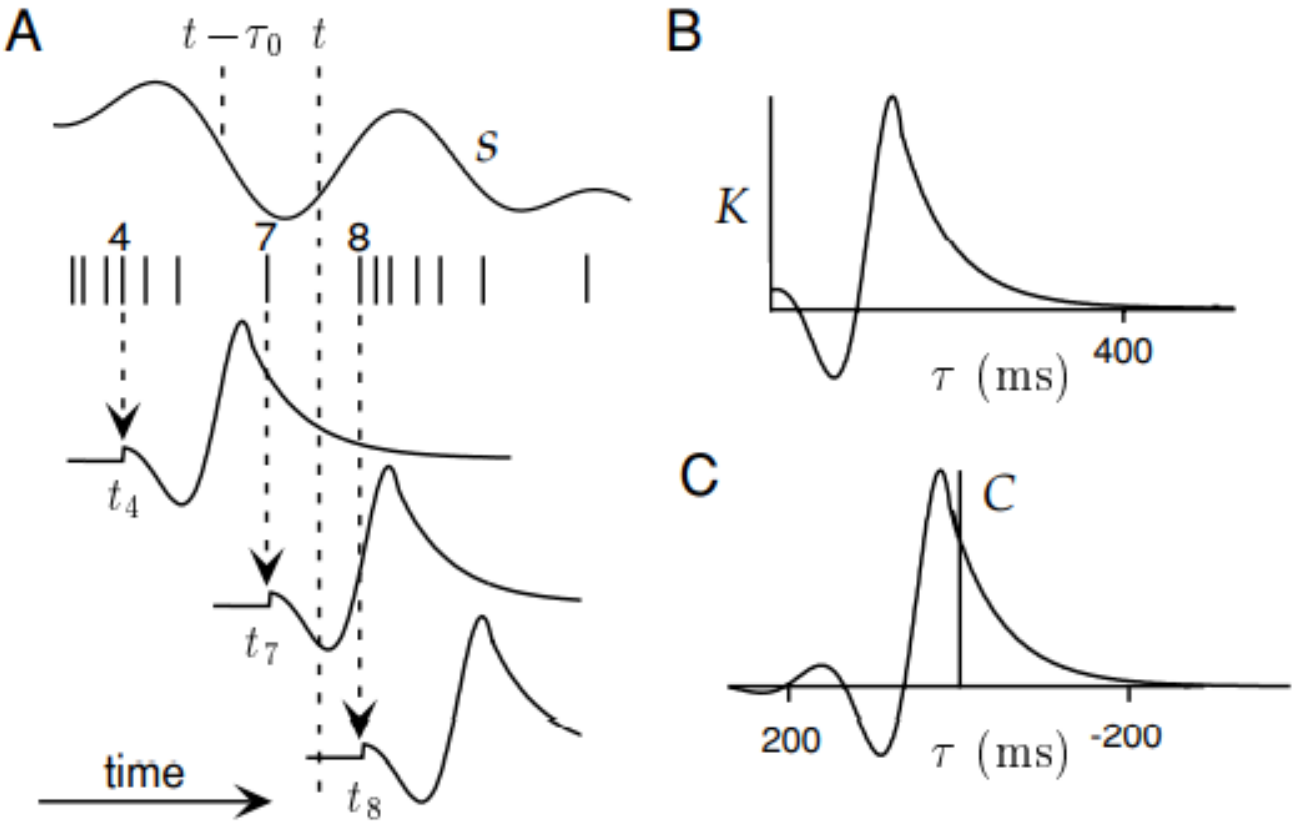
- We can estimate such a stimulus from the sequence of firing time of the spikes that it evokes.
- By approximation using kernel and sliding window function
- Assumption: time average of the stimulus being estimated is zero

- The firing of an action potential at time t_i is only affected by the stimulus $s(t)$ prior to that time $t < t_i$

-> Evoked spikes are telling about the past behavior of the stimulus

- In decoding, we want to extract information from the AP about the value of the stimulus at a later time $t > t_i$

-> By using past information, to predict the current stimulus value



- Tau_zero = prediction delay.
- Spikes ocrrurring prior to time t, estimate of the stimulus at time t - tau_zero

$$s_{\text{est}}(t-\tau_0) = \sum_{i=1}^{n} K(t-t_i) - \underline{\langle r\rangle \int_{-\infty}^{\infty} d\tau\, K(\tau)}\,.$$

- Stimulus estimate is a linear sum over all spikes(kernels)

To impose that time average of s_est is zero

the constraint is that K(t-t_i) = 0 for t_t_i <=0 : Causal (if kernel satisfies this condition)

$$s_{\text{est}}(t-\tau_0) = \int_{-\infty}^{\infty} d\tau\, (\rho(t-\tau) - \langle r\rangle)\, K(\tau)\,.$$

$$\frac{1}{T}\int_0^T dt \left\langle \left(\int_{-\infty}^{\infty} d\tau\, (\rho(t-\tau) - \langle r\rangle)\, K(\tau) - s(t-\tau_0) \right)^2 \right\rangle ,$$

$$\int_{-\infty}^{\infty} d\tau'\, Q_{\rho\rho}(\tau-\tau') K(\tau') = Q_{rs}(\tau-\tau_0)\,, \qquad (3.54)$$

where $Q_{\rho\rho}$ is the spike-train autocorrelation function,

$$Q_{\rho\rho}(\tau-\tau') = \frac{1}{T}\int_0^T dt\, \langle (\rho(t-\tau) - \langle r\rangle)(\rho(t-\tau') - \langle r\rangle)\rangle\,, \qquad (3.55)$$

Because we can completely control the stimulus

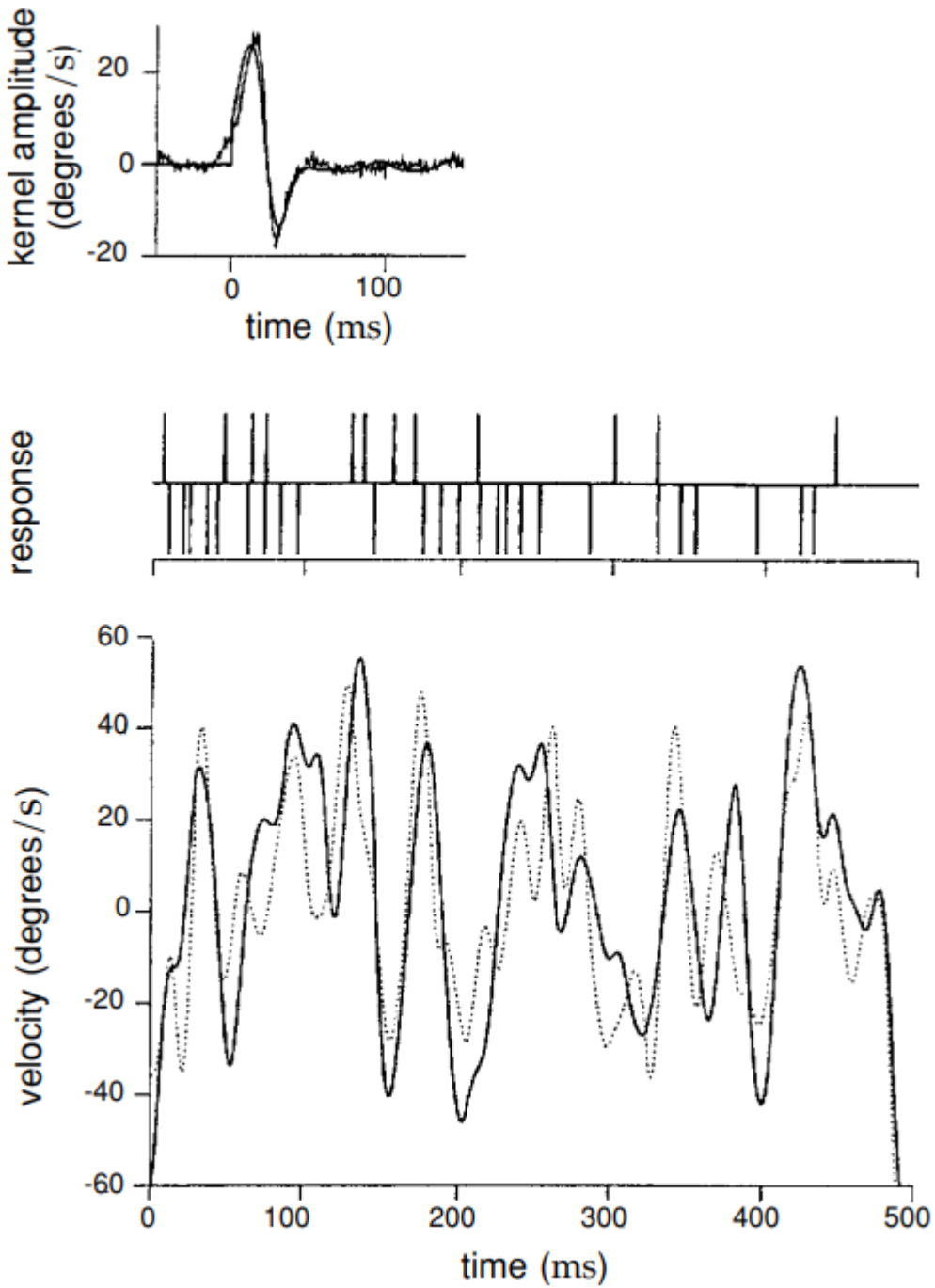
$$Q_{\rho\rho}(\tau) = \langle r \rangle \delta(\tau)\,, \qquad (3.57)$$

If spike train is uncorrelated

and we find from equation 3.54 that

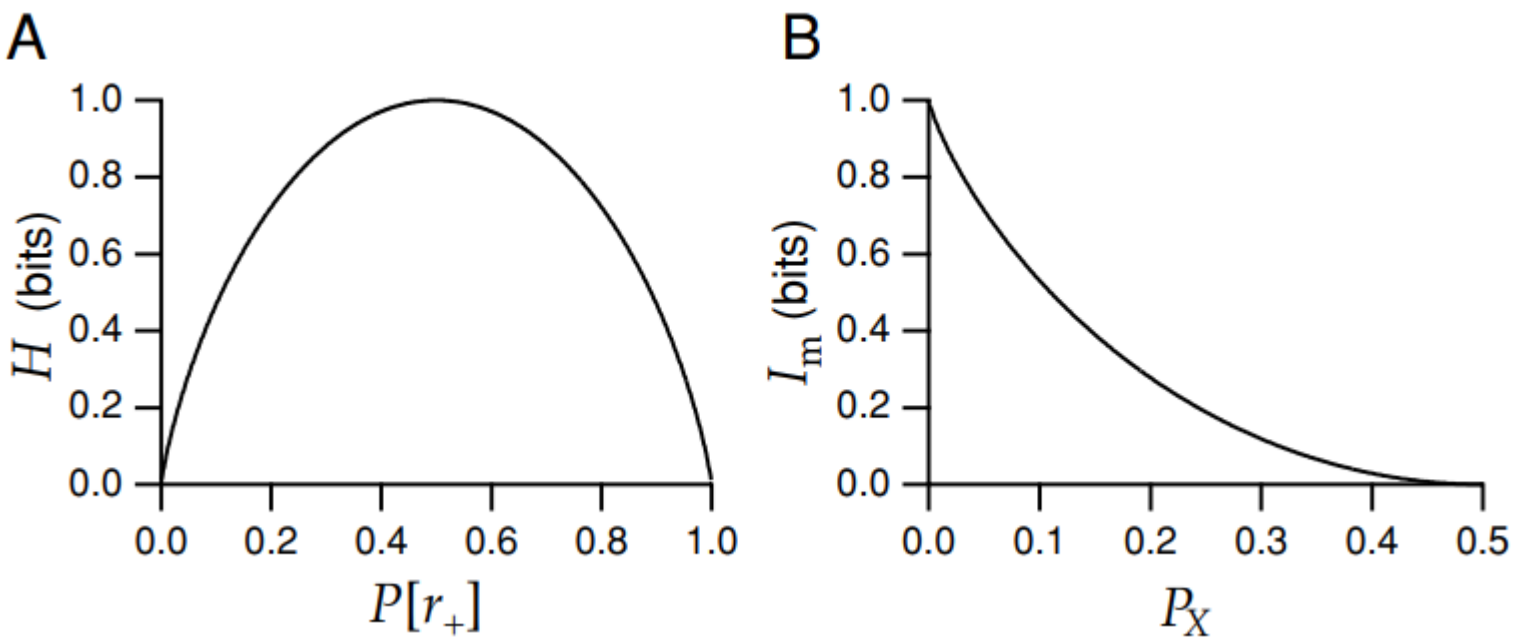
$$K(\tau) = \frac{1}{\langle r \rangle} Q_{rs}(\tau - \tau_0) = C(\tau_0 - \tau) = \frac{1}{\langle n \rangle} \left\langle \sum_{i=1}^{n} s(t_i + \tau - \tau_0) \right\rangle . \qquad (3.58)$$

- In causal system, correlations between a spike and subsequent stimuli can only arise from corr. Between the stim and itself.
- But if we ignore the constraint of causality, spike cannot depend on the behavior of a stimulus in the very distant past. by replacing $K(\tau)$ by $\Theta(\tau)K(\tau)$,



- Jagged: acausal
- Smooth: causal
- Dash: real
- Solid: estimated

Information Theory



$$h(P[r_1]P[r_2]) = h(P[r_1]) + h(P[r_2])\,.$$

$$h(P[r]) = -\log_2 P[r]\,.$$

$$H = -\sum_r P[r]\log_2 P[r]\,.$$

$$H_s = -\sum P[r|s]\log_2 P[r|s]\,.$$

$$H_{\text{noise}} = \sum_s P[s]H_s = -\sum_{s,r} P[s]P[r|s]\log_2 P[r|s]\,.$$

$$I_{\text{m}} = H - H_{\text{noise}} = -\sum_r P[r]\log_2 P[r] + \sum_{s,r} P[s]P[r|s]\log_2 P[r|s]\,.$$